Computer Vision

Homework 1: Binary Images on Paper

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In the spirit of Assignment 1 of CS 585 (Spring 2024), a paper-and-pencil assignment with no programming. Goes with Section 1.2 of the notes.

Write out complete arguments. $B(x, y)$ is a binary image, A its area, $(\bar{x}, \bar{y})$ its centroid and a, b, c its second central moments as defined in Section 1.2.4.

1. Show that the centroid is the unique point (x_0, y_0) minimizing $\sum_{x,y} \big[(x-x_0)^2+(y-y_0)^2\big] B(x, y)$, and that the minimum value is $a + c$.

2. Show that the central moments can be computed from the raw moments, $a = M_{20} - A\bar{x}^2$, $b = 2(M_{11} - A\bar{x}\bar{y})$ and $c = M_{02} - A\bar{y}^2$.

3. Starting from $E(\theta) = a\sin^2\theta - b\sin\theta\cos\theta + c\cos^2\theta$, show that the axis of least second moment satisfies $\tan 2\theta = b/(a - c)$, and explain why $\theta = \frac{1}{2}\operatorname{atan2}(b, a - c)$ always picks the minimum while $\frac{1}{2}\arctan\big(b/(a - c)\big)$ does not.

4. Compute a, b, c and the roundness $E_{\min}/E_{\max}$ for a filled axis-aligned rectangle of width w and height h (treat it as a continuous region). What happens when $w = h$, and why is a square "round" by this measure?

5. Prove that $E_{\min} \geq 0$ using the Cauchy–Schwarz inequality, and characterize the binary images for which $E_{\min} = 0$.

6. Show that the iterative threshold of Ridler and Calvard never increases the total squared error $J(\tau, \mu_1, \mu_2)$ of Section 1.2.1, and conclude that it stops after finitely many steps on an 8-bit image.

7. Let $D(d)$ count the object pixels with $x + y = d$. Prove that $\sum_d d^2 D(d) = M_{20} + 2M_{11} + M_{02}$, and find a fourth projection that would give M_{11} on its own when combined with D.